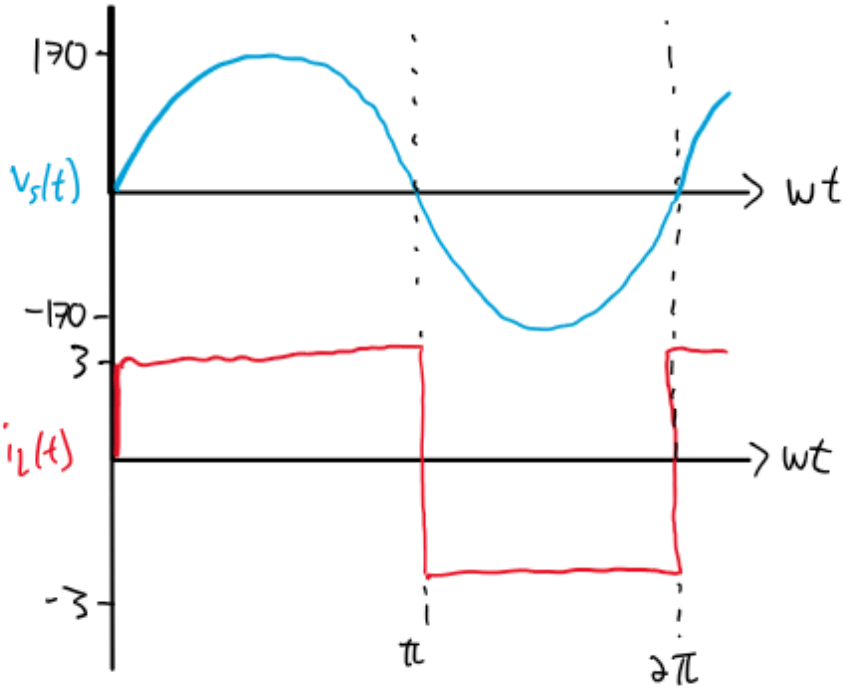
A3 : Distorted current from a sinusoidal source



a) For square-wave AC signal: $\hat{V}_k = \frac{4 \cdot V_{DC}}{k \cdot \pi} \cdot \left| \sin\left(k \cdot \frac{\pi}{2}\right) \right|$

↑ absolute value

$$\hat{I}_{,1} = \frac{4}{\pi} \cdot I_{DC} = \frac{4}{\pi} \cdot 3A = \frac{12}{\pi} A$$

$$\boxed{I_{1,rms} = \frac{12}{\pi\sqrt{2}} = 2.7A} \quad \therefore \quad I_{1,rms\ \text{square wave}} = 0.9 \cdot I_{DC}$$

$$THD = \frac{\sqrt{I_{rms}^2 - I_{1,rms}^2}}{I_{1,rms}} = \frac{\sqrt{3^2 - 2.7^2}}{2.7} = \boxed{48.3\%}$$

Average power (long way)

$$\langle P \rangle = \frac{1}{2\pi} \int_0^{2\pi} v(\omega t) \cdot i(\omega t)\, d\omega t$$

$$= \frac{1}{2\pi} \cdot \left[\int_0^{\pi} 170\sin(\omega t) \cdot 3 \, d\omega t + \int_{\pi}^{2\pi} 170\sin(\omega t) \cdot (-3)\, d\omega t \right]$$

$$= \frac{510}{2\pi} \cdot \left[\int_0^{\pi} \sin(\omega t)\, d\omega t - \int_{\pi}^{2\pi} \sin(\omega t)\, d\omega t \right]$$

$$= \frac{510}{2\pi} \cdot \left[-\cos(\pi) + \cos(0) + \cos(2\pi) - \cos(\pi) \right]$$

$$= \frac{510}{2\pi} \cdot \left[1 + 1 + 1 + 1 \right]$$

$$= \frac{1020}{\pi} W = \boxed{325 W}$$

Average power (short-cut)

we know $v(t)$ and $i(t)$ are

in-phase $\therefore$ apparent power = real power

$$S = P$$

$$S_1 = V_{1,rms} \cdot I_{1,rms} = \frac{170}{\sqrt{2}} V \cdot 2.7A = 325 W$$

b) Displacement factor = $\cos\theta$ $\quad \theta = \theta_v - \theta_I$

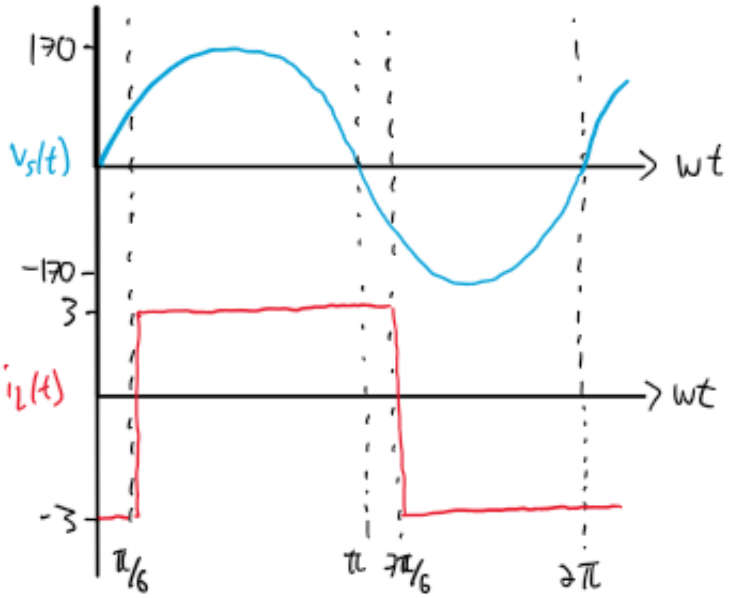
$\theta_v = \theta_I$ when zero-crossings match ∴

displacement factor = $\boxed{1}$

Distortion factor = $\dfrac{I_{1,rms}}{I_{rms}} = \dfrac{2.7}{3} = \boxed{0.9}$

Power factor : k_p = displacement · distortion = $1 \cdot 0.9 = \boxed{0.9}$

c)



Average power (long-way)

- Do the integral for $\langle p \rangle$ but there will be 4 sections instead of 2.

Average power (short-way)

displacement factor: $\cos\theta = \cos(\theta_v - \theta_I) = \cos(0 - 30°) = 0.866$

distortion factor: unchanged $\therefore 0.9$

$k_p = 0.866 \cdot 0.9 = \boxed{0.78}$

$\langle p \rangle = S \cdot k_p = V_{rms} \cdot I_{rms} \cdot k_p = \frac{170}{\sqrt{2}} \cdot 3 \cdot 0.78 = \boxed{281\ W}$

↑ apparent power

or

$\langle p \rangle = V_{1,rms} \cdot I_{1,rms} \cdot \cos\theta$

↑ assuming no voltage harmonics

Kassakian 8.4: Removing a harmonic with phase-shift control

Why bother using complex gating patterns to remove harmonics? Can't we use filters?

$$\frac{\hat{V}_k}{V_{oc}} = \frac{4}{k\pi} \cdot \left|\cos(k\pi a)\right| \cdot \underbrace{\left|\sin\left(k \cdot \frac{\pi}{2}\right)\right|}_{\text{odd harmonics only}}$$

($\hat{V}_k$: peak)

to remove the 7th harmonic, $\hat{V}_7 = 0$

$$\cos(7\pi a) = 0$$

$$7\pi a = \frac{\pi}{2} \text{ or } \frac{3\pi}{2} \text{ or } \frac{5\pi}{2} \text{ or } \frac{7\pi}{2}$$

$$a = \frac{1}{14} \text{ or } \frac{3}{14} \text{ or } \frac{5}{14} \text{ or } \frac{7}{14}$$

$a < 0.5$ or there is no waveform

$\therefore$

$$a = \frac{1}{14} \text{ or } \frac{3}{14} \text{ or } \frac{5}{14}$$

Which one should we pick?

Answer: Whatever gives us the largest fundamental frequency component because that's what we want and everything else is in the way.

Intuitively, that means whatever spends the least amount of time in the zero-voltage state.

$\therefore a_{best} = \frac{1}{14}$

Check impact on fundamental, 3rd and 5th :

$$\hat{V}_1 = \frac{4}{\pi} \cdot \cos\left(\frac{\pi}{14}\right) = \frac{4}{\pi} \cdot 0.975$$

↓

2.5% reduction : not bad

$$\hat{V}_3 = \frac{4}{3\pi} \cdot \cos\left(\frac{3\pi}{14}\right) = \frac{4}{3\pi} \cdot 0.782$$

↓

21.8% reduction : nice

$$\hat{V}_5 = \frac{4}{5\pi} \cdot \cos\left(\frac{5\pi}{14}\right) = \frac{4}{5\pi} \cdot 0.434$$

↓

56.6% reduction : nice

if we chose $a = \frac{5}{14}$,

$$\hat{V}_1 = \frac{4}{\pi} \cos\left(\frac{5\pi}{14}\right) = \frac{4}{\pi} \cdot 0.434$$

↓

56.6% reduction : bad

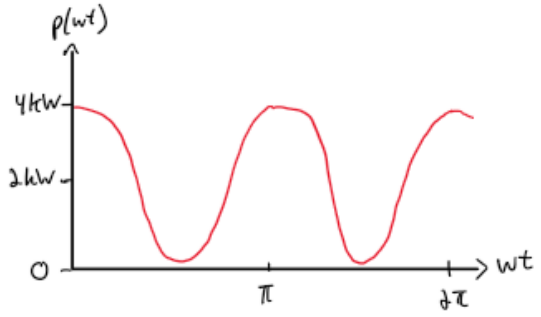
and

$$\hat{V}_3 = \frac{4}{3\pi} \cos\left(\frac{15\pi}{14}\right) = \frac{4}{3\pi} \cdot -0.975$$

↓

only 2.5% reduction : bad

C1 : Following the power to the DC side



Assume $v(t)$ & $i(t)$ are $\cos(\omega t)$ signals, although they could be $\sin(\omega t)$ as well.

a) Trig identity for $\hat{V}\cos(\omega t)\cdot\hat{I}\cos(\omega t) = \frac{\hat{V}\hat{I}}{2}\cdot\left(1+\cos(2\omega t)\right)$

↑ Average power @ unity

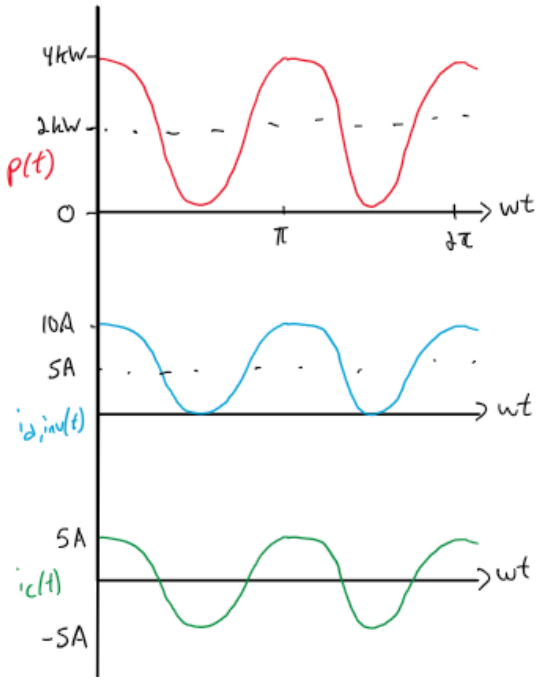
$$p(t) = 2000 + 2000\cos(2\omega t)\,W$$

b) $I_d = \frac{\langle P\rangle}{V_d} \quad \frac{2000}{400} = 5A$

$i_c = 5\cos(2\omega t)$ ← match instantaneous power phase

∴ if $p(t) > P$, $i_c > 0$

$i_{d,inv} = I_d + i_c = 5 + 5\cos(2\omega t)$



c) Find the DC capacitor size to keep $V_d(t)$ within 1% of 400V

$V_{d\,min} = 396V$ $\quad \Delta V_d = 8V$

$V_{d\,max} = 404V$

$$\Delta E_{cap} = \int_0^{\pi/2w} P\sin(2wt)\,dt = \frac{P}{2w}\cdot\left[-\cos(\pi)+\cos(0)\right] = \frac{P}{w}$$

$$C \geq \frac{\Delta E}{V_d \cdot \Delta V_d} \qquad C \geq \frac{P}{w V_d \cdot \Delta V_d}$$

$f = 60Hz \therefore w = 120\pi$ rad/s

$$C \geq \frac{2000}{120\pi \cdot 400 \cdot 8} = 1.66mF$$

d) What's the ripple if $C = 1mF$?

$$\Delta V_d = \frac{\Delta E}{C \cdot V_d} = \frac{P}{w \cdot C \cdot V_d} = \frac{2000}{120\pi \cdot 10^{-3} \cdot 400} = 13.3V$$

$\frac{13.3V}{400V} = 3.3\%$ peak-to-peak ripple

e) Load changes, power factor is now 0.8 lagging but P is still 2kW

P remains constant, so does I_d $\therefore$ $I_d = 5A$

Apparent power: $\frac{P}{k_p} = \frac{2000}{0.8} = 2.5kVA$

$\Delta E = \frac{S}{\omega}$ $\therefore$ $C \geq \frac{2500}{120\pi \cdot 400 \cdot 8} = 2.07mF$

f) Why is the capacitor size inversely proportional to V_d and ΔV_d?
Does raising the switching frequency help?

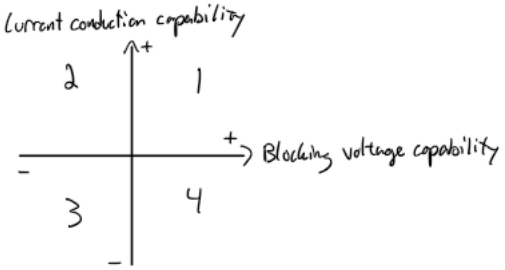
Answer:

Only the line frequency matters because that sets how long the capacitor must store or release energy for ($p(t) > P$ or $p(t) < P$).

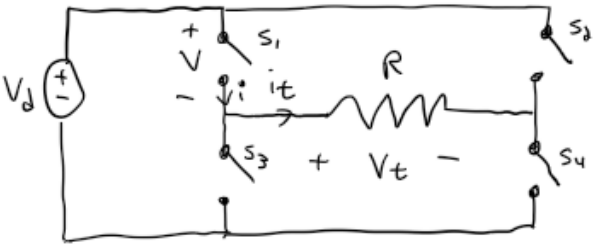
$\Delta E = C \cdot V_d \cdot \Delta V_d$

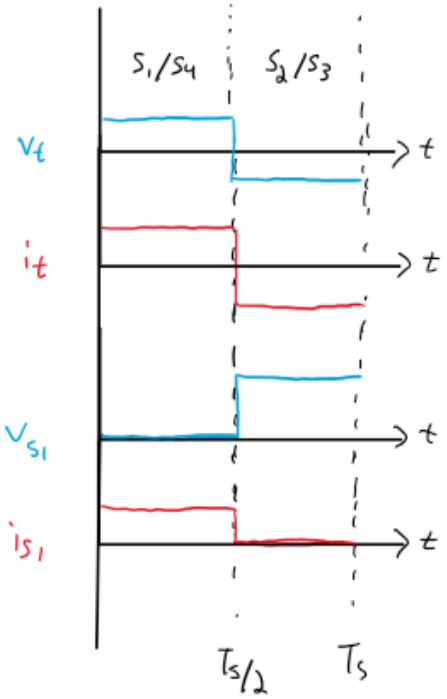
for the same amount of energy buffered $\left(\frac{S}{\omega}\right)$, either the voltage needs to be high or the capacitance does to keep the same amount of ripple.

D1 : Which quadrants?



a) Find quadrants and a switching device

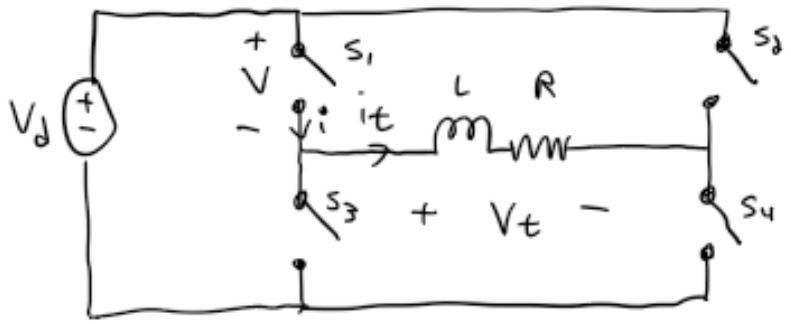


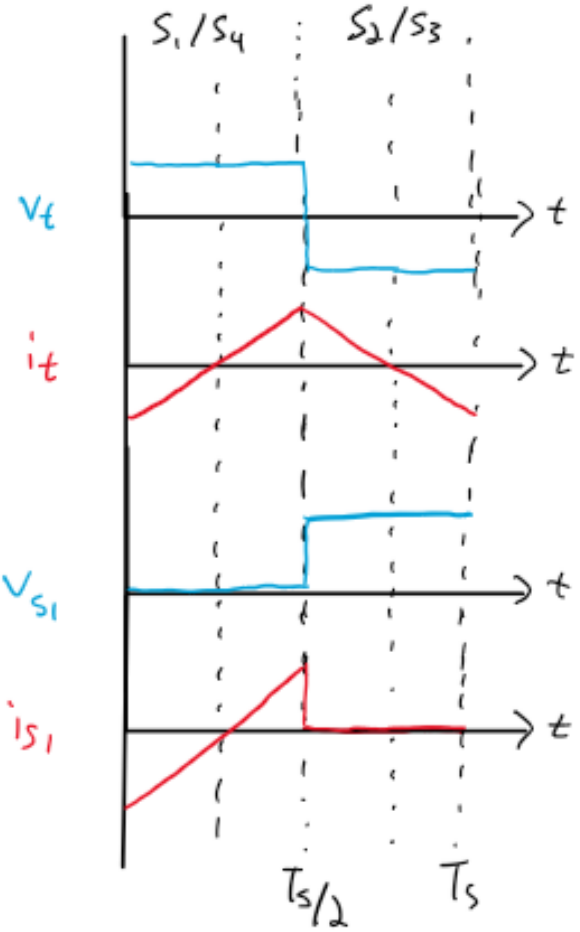


Quadrant 1 : positive voltage blocking
positive current conduction

device : IGBT (MOSFET would work)

b) Still square wave

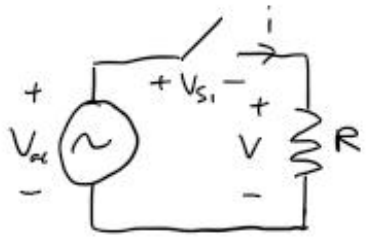




Quadrants: 1 and 4
Both current directions,
Blocking positive voltages

Device: MOSFET or IGBT w/ parallel diode

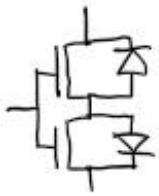
c)



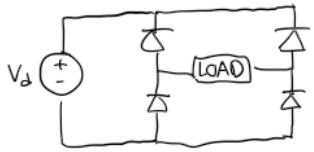
Needs to turn on or off
at any time in AC waveform

All four quadrants.

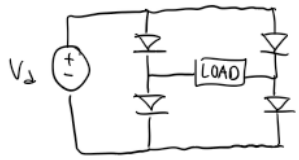
Device : back-to-back MOSFETs
or a bidirectional switch (new device)



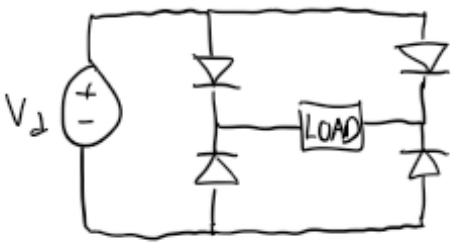
d) Could you only use diodes for the full bridge?



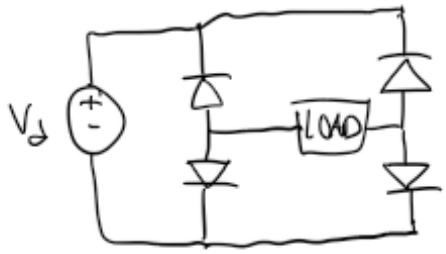
Result : no current to the load



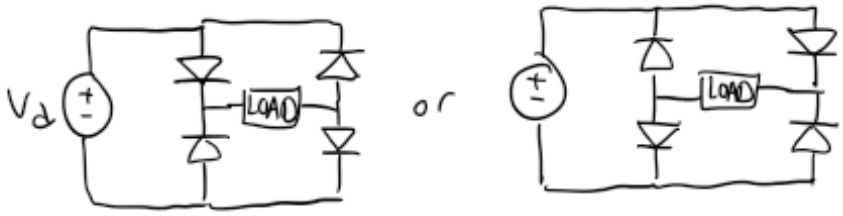
Result : short-circuit of the DC voltage



Result: Zero-voltage applied to the load



Result: no current to the load



Result: V_d applied to load as a DC voltage.

Might as well go: 

and remove the diodes.

∴ No. You need a switch that can be turned on.